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Experiment 01

Aim:Version control operations using Git.

This experiment maps LO2

LO2: To understand and perform the basic version control operations using Git and GitHub, including repository creation, cloning, file tracking, committing, pushing, pulling, branching, merging, and remote repository management.

Theory: Git is a distributed version control system that helps developers track changes in source code and files. It enables efficient project management through repositories, commits, branches, and remote repositories, making collaboration and version management easier. In this experiment, various Git commands are performed to understand the basic operations of version control.

Command	Syntax	Purpose
Initialize Repository	git init	Creates a new Git repository.
Version Check	git version--	Checks the git version
Check Status	git status	Shows modified, staged, and untracked files.
Master Branch	git branch master -m main	Changes Master branch name to main
Stage All Files	git add .	Adds all changes to the staging area.
Stage One File	git add python.py	Adds only python.py to staging.
Commit Changes	git commit -m "message"	Saves staged changes with a message.
View Commit History	git log	Shows complete commit history.
One-line History	git log --oneline	Shows compact commit history.
Push to GitHub	git push origin main	Uploads local commits to the remote main branch.
Clone Repository	git clone <url>	Copies a GitHub repository to your computer.
List Branches	git branch	Displays all local branches.

Command	Syntax	Purpose
Switch Branch	git checkout main	Switches to the main branch.
Create & Switch	git checkout -b arvind	Creates and switches to a new branch.
Push Branch	git push origin arvind	Pushes the arvind branch to GitHub.
Pull Changes	git pull origin main	Fetches and merges changes from the remote main.
Fetch Only	git fetch origin	Downloads remote updates without merging them.
Merge Branch	git merge arvind	Merges arvind into the current branch.
Restore File	git restore python.py	Discards uncommitted changes in python.py.
Restore All Files	git restore .	Discards all uncommitted changes.
Unstage File	git restore --staged python.py	Removes python.py from the staging area.
Soft Reset	git reset --soft HEAD~1	Removes the last commit but keeps changes staged.
Hard Reset	git reset --hard HEAD~1	Removes the last commit and discards all changes.
Reset to Remote	git reset --hard origin/main	Makes your local branch exactly match origin/main.
Show Differences	git diff	Displays unstaged changes.
Show Staged Differences	git diff --staged	Displays staged changes.
Show Remote	git remote -v	Lists configured remote repositories.
Add Remote	git remote add origin <url>	Connects a local repository to GitHub.
Delete Branch	git branch -d arvind	Deletes the local arvind branch.
Delete Remote Branch	git push origin --delete arvind	Deletes the remote arvind branch.
Force Push Safely	git push --force-with-lease origin main	Rewrites the remote branch after local history changes, with a safety check.
Set Username	git config --global user.name "Arvind-9206"	Sets your Git username globally for all repositories.
Set Email	git config --global user.email	Sets your Git email globally for all repositories.

Command	Syntax	Purpose
	"arvindpatil.9206@gmail.com"	

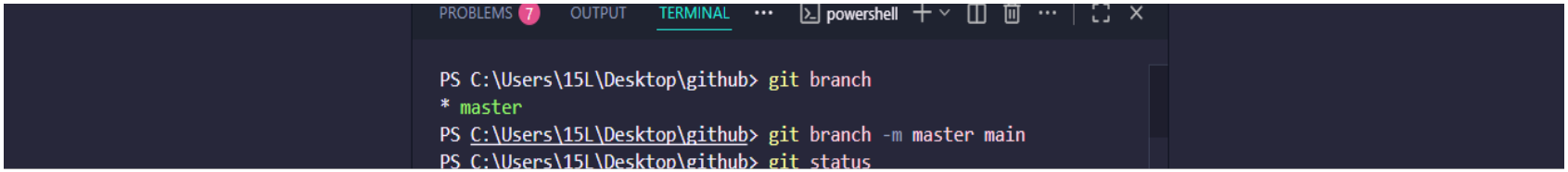
Output

```
PS C:\Users\15L\Desktop\github> git init
Initialized empty Git repository in C:/Users/15L/Desktop/github/.git/
```

```
C:\Users\15L>git --version
git version 2.47.1.windows.2
```

```
@Arvind-2906 →/workspaces/Game (main) $ git status
On branch main
Your branch is up to date with 'origin/main'.

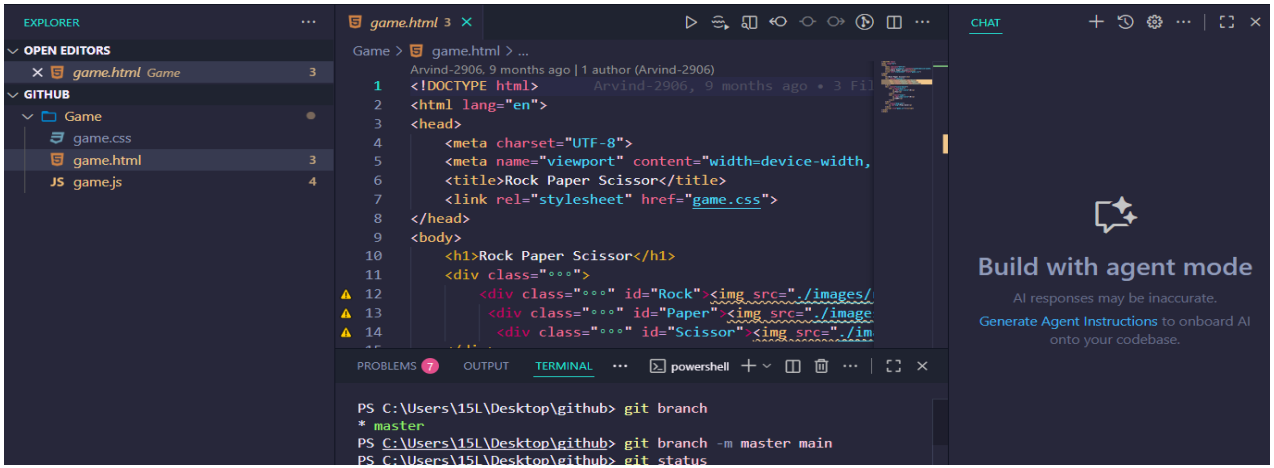
nothing to commit, working tree clean
```

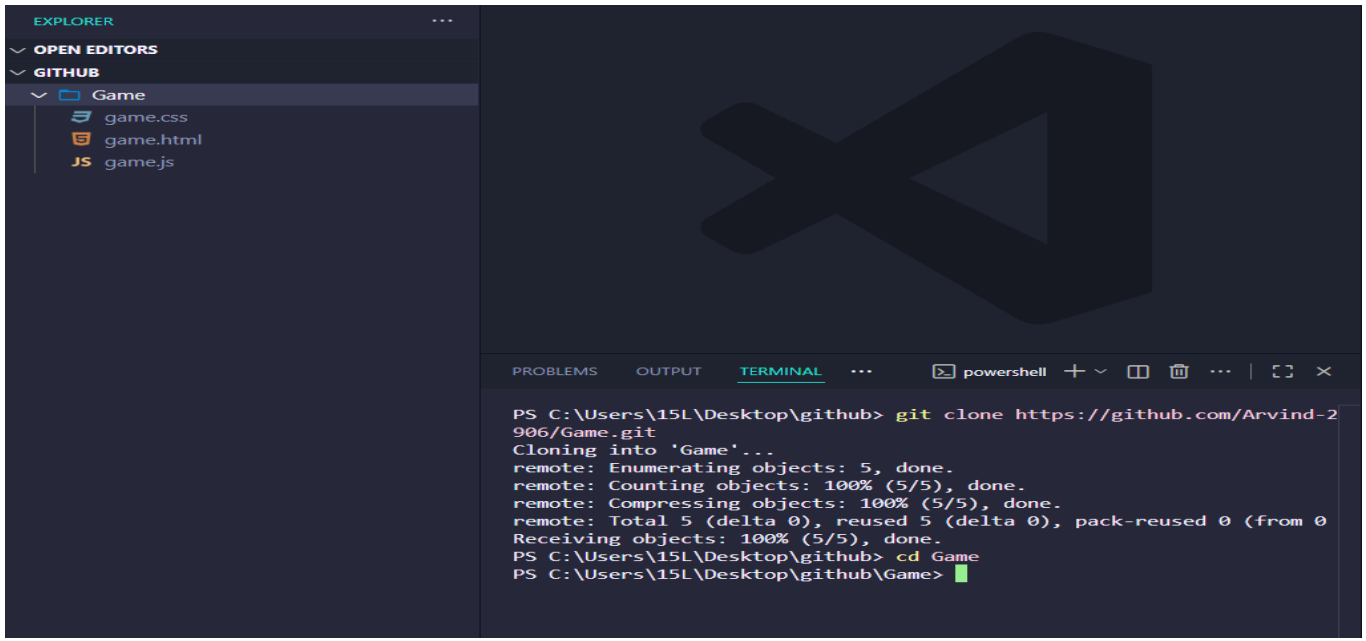


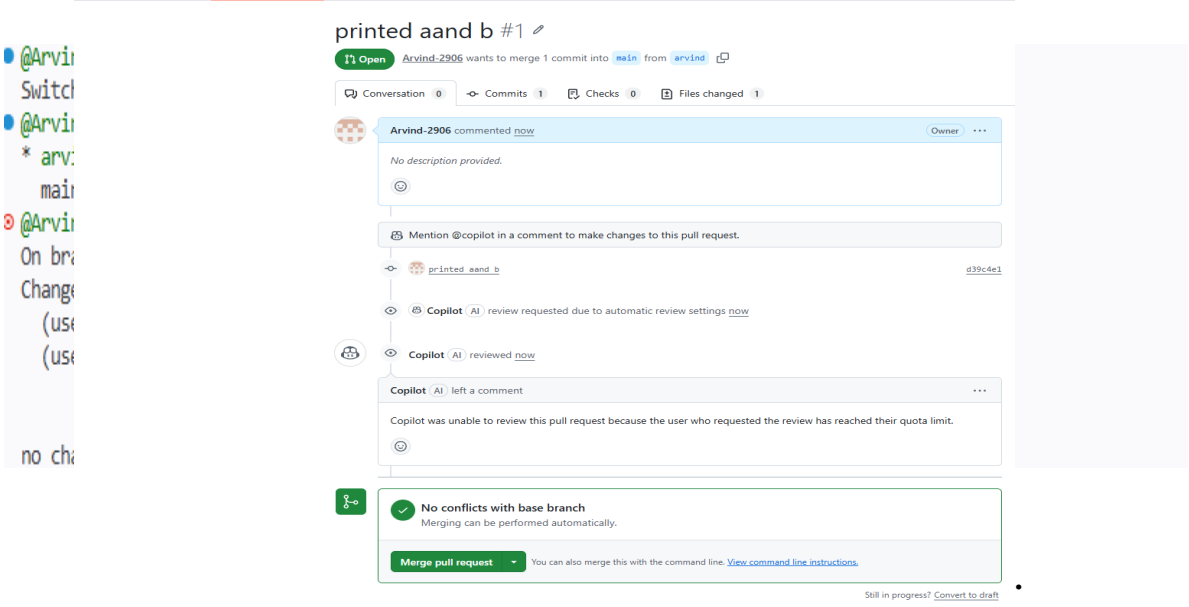
```
@Arvind-2906 →/workspaces/Game (arvind) $ git add .
@Arvind-2906 →/workspaces/Game (arvind) $ git commit -m "printed aand b"
[arvind d39c4e1] printed aand b
 1 file changed, 3 insertions(+), 1 deletion(-)
```

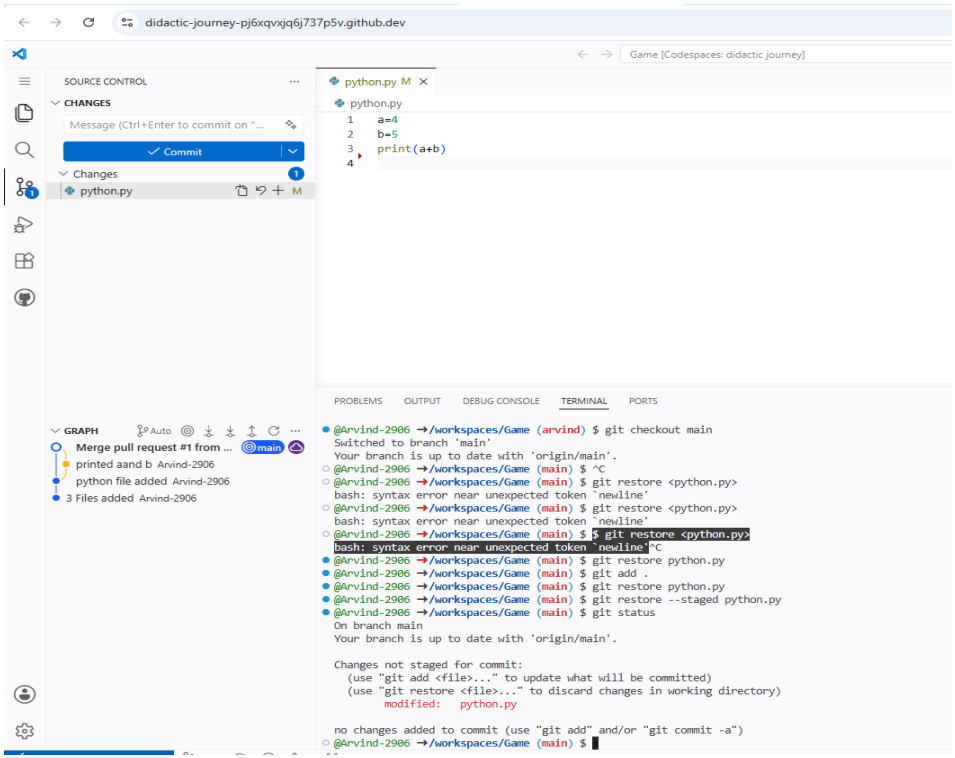
```
PS C:\Users\15L\git_lab_1> git log --oneline
f040f58 (HEAD -> master, feature-branch) Added feature.txt
6a769a4 Added experiment.txt file
```

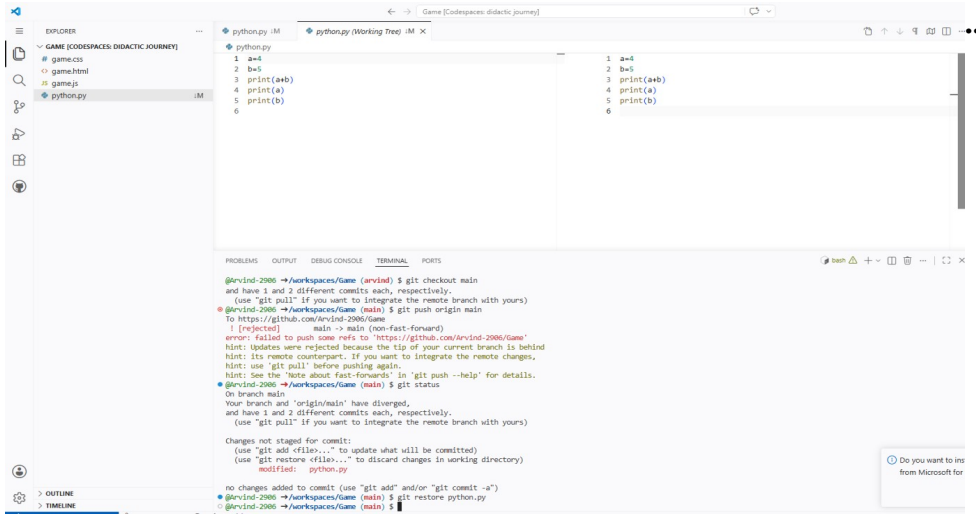
```
@Arvind-2906 →/workspaces/Game (main) $ git push origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 294 bytes | 294.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/Arvind-2906/Game
   462227d..4931fc0  main -> main
```




EXPLORER
OPEN EDITORS
GITHUB
Game
game.css
game.html
game.js
PROBLEMS
OUTPUT
TERMINAL
powershell
PS C:\Users\15L\Desktop\github> git clone https://github.com/Arvind-2906/Game.git
Cloning into 'Game'...
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (5/5), done.
remote: Total 5 (delta 0), reused 5 (delta 0), pack-reused 0 (from 0
Receiving objects: 100% (5/5), done.
PS C:\Users\15L\Desktop\github> cd Game
PS C:\Users\15L\Desktop\github\Game>


printed aand b #1
Open
Arvind-2906 wants to merge 1 commit into main from arvind
Conversation 0
Commits 1
Checks 0
Files changed 1
Arvind-2906 commented now
Owner
No description provided.
Mention @copilot in a comment to make changes to this pull request.
printed aand b
d39c4e1
Copilot AI review requested due to automatic review settings now
Copilot AI reviewed now
Copilot AI left a comment
Copilot was unable to review this pull request because the user who requested the review has reached their quota limit.
No conflicts with base branch
Merging can be performed automatically.
Merge pull request
You can also merge this with the command line. View command line instructions.
Still in progress? Convert to draft


didactic-journey-pj6xqvxjq6j737p5v.github.dev
Game [Codespaces: didactic journey]
SOURCE CONTROL
CHANGES
Message (Ctrl+Enter to commit on "...
Commit
Changes
python.py
M
python.py M
python.py
1 a=4
2 b=5
3 print(a+b)
4
GRAPH
Auto
Merge pull request #1 from ...
main
printed aand b Arvind-2906
python file added Arvind-2906
3 Files added Arvind-2906
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
@Arvind-2906 →/workspaces/Game (arvind) $ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
@Arvind-2906 →/workspaces/Game (main) $ ^C
@Arvind-2906 →/workspaces/Game (main) $ git restore <python.py>
bash: syntax error near unexpected token `newline'
@Arvind-2906 →/workspaces/Game (main) $ git restore <python.py>
bash: syntax error near unexpected token `newline'
@Arvind-2906 →/workspaces/Game (main) $ $ git restore <python.py>
bash: syntax error near unexpected token `newline'^C
@Arvind-2906 →/workspaces/Game (main) $ git restore python.py
@Arvind-2906 →/workspaces/Game (main) $ git add .
@Arvind-2906 →/workspaces/Game (main) $ git restore python.py
@Arvind-2906 →/workspaces/Game (main) $ git restore --staged python.py
@Arvind-2906 →/workspaces/Game (main) $ git status
On branch main
Your branch is up to date with 'origin/main'.
Changes not staged for commit:
(use "git add <file>..." to update what will be committed)
(use "git restore <file>..." to discard changes in working directory)
modified: python.py
no changes added to commit (use "git add" and/or "git commit -a")
@Arvind-2906 →/workspaces/Game (main) $




```
@Arvind-2906 ➜/workspaces/Game (main) $ git push --force-with-lease origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 281 bytes | 281.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/Arvind-2906/Game
 + 6eb83c7...09c7e81 main -> main (forced update)
```

,

```
PS C:\Users\15L\git_lab_1> git branch feature-branch
PS C:\Users\15L\git_lab_1> git switch feature-branch
Switched to branch 'feature-branch'
```

..

```
PS C:\Users\15L\git_experiment_20> echo "Version 2.0 update" >> app.txt
PS C:\Users\15L\git_experiment_20> git diff
diff --git a/app.txt b/app.txt
index 3a62f4c..1125776 100644
Binary files a/app.txt and b/app.txt differ
```

CONCLUSION

The experiment demonstrated the use of essential Git version control commands for repository management, tracking changes, branch handling, and collaboration with GitHub. It provided practical knowledge of managing and maintaining software projects efficiently using Git.